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Review article



# Key operational considerations for biogeographical ancestry prediction capabilities

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ABSTRACT

Biogeographical ancestry (BGA) prediction is a valuable Forensic DNA Intelligence (FDI) tool used to provide investigators with additional information about the donor of biological evidence. FDI is particularly useful in cases where routine DNA analysis has not led to an identification. Over the last two decades, substantial research effort has been directed towards the development of new panels of ancestry informative markers and analysis tools for BGA prediction. These efforts have supported the successful application of BGA prediction in casework across several international jurisdictions. This paper identifies critical components of operational BGA prediction methods to highlight considerations for method development, optimisation, validation and associated interpretation and to identify areas where further research is needed to enhance capabilities.

## 1. Introduction

Law Enforcement (LE) agencies are adopting intelligence-led models of policing in which forensic analyses are utilised to assist with generating investigative leads [1–4]. Forensic DNA Intelligence (FDI) tools, encompassing the prediction of biogeographical ancestry (BGA) and Forensic DNA Phenotyping, assist investigations by providing additional information about the donor of biological evidence. These tools are particularly valuable where routine DNA analysis, using short tandem repeats (STRs), has failed to lead to an identification and can be useful to: i) support or refute eyewitness statements, ii) assist in the prioritisation of suspect pools or lists of missing persons, and iii) support requests to conduct DNA screening activities to provide suitable references for routine DNA analysis using STRs.

BGA prediction has become an important FDI capability through continued focus on technological development, forensic validation and operationalisation. Early research by Shriver *et al.* [5] was foundational to the development of BGA prediction assays and supported the first operational capability established by DNAPrint Genomics in the early 2000s [6–9]. Assays for BGA prediction have since been expanded to increase the number of informative markers and novel marker types [10–16]. This development was facilitated by the advancement of sequencing technologies enabling targeted and whole genome sequencing (WGS) along with large-scale data analysis capabilities [17,

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18]. These advancements have supported the continued operational use of BGA prediction methods in several international jurisdictions [19–22].

The next step in advancing operational capabilities should focus on identifying strategies to improve practices for method optimisation, validation, interpretation and reporting of BGA predictions. This work would be valuable to support greater consistency in the validation and application of BGA prediction methods in forensic investigations. Appropriate validation is essential to ensure the performance characteristics and limitations of a forensic capability are understood [23,24]. To date, BGA prediction capabilities have been validated following guidelines developed for STR analysis method which do not account for the unique features of BGA prediction capabilities. For BGA prediction methods, validation studies targeted at answering the following questions are needed: i) under which circumstances will a method provide a reliable prediction? ii) when might a result be misleading or erroneous? iii) can admixture be detected and accurately resolved? iv) how can uncertainty inherent to BGA predictions be captured in the interpretation and reporting of results? The answers to these questions will depend on the characteristics of the specific methods used, including the ancestry informative marker (AIM) panel/s, reference population data, and classification tools. A thorough understanding of the performance characteristics of these components is critical to enable operational laboratories to develop appropriate strategies for interpretation and reporting that reflect the capacity and limitations of the specific method in use.

This paper discusses the challenges and considerations associated with the operational implementation of BGA prediction capabilities with a focus on method optimisation, validation, results interpretation and reporting and highlights areas where further research is required to inform best practice.

## 2. Key components of BGA prediction methods

Essential to the development of an operational BGA prediction capability is the prior understanding of the characteristics and limitations of three key components: i) the AIM panel/s used, ii) the reference population data, and iii) the classification algorithms and analysis tools.

### 2.1. Ancestry informative marker panels

BGA prediction relies on the analysis of AIMs, which are characterised by allele frequencies that differ significantly between populations of interest (POI) but which remain stable within a population [5,25,26]. The informativeness of AIMs depend upon the measured allele frequency differentials between POI [27,28]. Defining the POI is therefore an important consideration when selecting an AIM panel for operational use. Generally, operational forensic laboratories seeking to implement BGA prediction capabilities are unlikely to have the resources and expertise required to develop a custom AIM panel for the POI relevant to their jurisdiction. As a result, laboratories tend to have a preference to validate and implement an existing published panel that will meet operational needs. Numerous single nucleotide polymorphisms (SNPs), including binary and multi-allelic SNPs, and microhaplotype markers (multiple SNPs closely located on the genome [29]), have been identified as being highly informative for BGA and form the basis of most AIM panels developed for forensic applications [18]. Insertion-deletion (InDel) markers are another class of AIMs that have been incorporated into AIM panels for operational use [30,31].

Many of the first published forensic AIM panels were first developed for use with capillary electrophoresis (CE) technology, readily available in forensic laboratories. SNP genotyping strategies using CE have predominantly relied on single base extension minisequencing assays using SNaPshot® chemistry [32,33], whereas a PCR-to-CE method (more analogous to routine STR typing strategies) has been preferred for panels of InDels [30,31,34]. Among the SNaPshot® panels developed for operational use were those designed for global population differentiation, such as the SNP*for*ID panel [35–37], and several panels for regional differentiation, including the Eurasiaplex [38] and Pacifiplex [33] panels.

Operational capabilities have continued to develop with technological advancements and AIM panels designed for massively parallel sequencing (MPS) technology have since become the focus for operational use [18,39]. MPS has enabled efficient genotyping of hundreds of markers in a single assay aiding the use of expanded AIM sets when compared to CE. MPS allows for multiple AIM types to be incorporated in a single analysis to increase differentiation between POI [18]. Table 1 summarises AIM sets developed for forensic applications using MPS technology.

Most forensic AIM panels for MPS have been designed for global differentiation of major continental populations [9,15,16,21,40–43]. However, several AIM panels have been developed to allow improved differentiation of regional populations, including MAPlex (Multiplex for the Asia-Pacific) [11] and the EUROFORGEN NAME (North African and Middle East) panel [44]. The global and regional panels developed for MPS incorporate many of the highly informative markers originally included in the panels developed using SNaPshot® technology [33, 35–38].

Published comparative performance assessments between several of these panels demonstrate similar performance for inter-continental population differentiation. A comparison of the EUROFORGEN gAIMs, ForenSeq Signature Prep Kit (SPK), MAPlex and Precision ID Ancestry Panel (PIAP) panels by Alladio *et al.* [55] demonstrated that all had high performance for the differentiation African, European and East Asian populations (AUC-ROC>0.99) with lower differentiation of Middle East, Central Asia, South Asia, Native America, Oceania (AUC-ROC 0.60 – 0.80). Differentiation of South Asian and Native American populations were the main differences between panels. Using MAPlex, South Asian differentiation was higher (AUC-ROC >0.80) compared to all other panels (AUC-ROC 0.60–0.71), however, differentiation of Native American populations was lower (AUC-ROC = 0.77) compared to the other panels tested (AUC-ROC >0.95). These performance characteristics reflect the MAPlex panel design which focused on AIMs to differentiate Asian-Pacific populations such as South Asia. Performance metrics for all panels decreased for intra-continental differentiation as expected given the difficulties associated with differentiation of closely related populations. Resutik *et al.* [57] reported similar findings in a comparative assessment of PIAP, MAPlex and the VISAGE Basic Tool (BT) with respect to differentiation of major continental populations; African, European and East Asian. This study also demonstrated that these three assays, each with less than 250 AIMs, performed similarly to a set of 10,000 genome wide SNPs highlighting the power of carefully selected minimal AIM panels with high informativeness across POI.

### 2.2. Reference population data

Accurate BGA prediction relies on the availability of reference population samples for relevant global and regional populations. Data from these samples are used in the discovery of AIMs and to assess their informativeness in different populations, as well as for the training of classification algorithms used to generate BGA predictions [13,56].

Several publicly accessible human genomic databases comprising data from individuals across diverse global populations have been used to support operational BGA prediction capabilities. These include the 1000 Genomes Project [57,58], Centre d'Etude du Polymorphisme Humain Human Genome Diversity Project (HGDP-CEPH) [59,60], Estonian Biocentre Genome Diversity Panel (EGDP) [61], and Simons Foundation Human Genome Diversity Project (SGDP) [62] as shown in Table 2. Reference data sets used operationally to train classification algorithms are typically comprised of data from these publicly available databases [63] with in-house generated data representing relevant local or regional populations incorporated, where possible. However, limited

**Table 1**
Summary of ancestry informative marker (AIM) panels evaluated for operational use with Massively Parallel Sequencing technology.

| AIM Set | Target | Populations Differentiated | Number & Type of AIMs* | Reference (s) |
|---|---|---|---|---|
| EUROFORGEN Global AIMs | Autosomal | African, East Asian, European, Native American, Oceanian | 128 binary SNPs | [14,45] |
| EUROFORGEN NAME | Autosomal | North African, Middle Eastern | 111 binary SNPs | [44,46] |
| FORCE** | Autosomal | African, Native American, East Asian, European, Oceanian, South Asian | 237 binary SNPs | [47] |
| ForenSeq Signature Prep Kit*** | Autosomal | African, East Asian, European, Admixed American | 56 binary SNPs | [48,49] |
| MAPlex | Autosomal | African, Native American, East Asian, European, Oceanian, South Asian | 108 binary SNPs<br>36 multi-allelic SNPs<br>20 microhaplotypes | [11,50] |
| Phenotrivium# | Autosomal<br>Y-Chromosome | African, Native American, East Asian, European, Oceanian, South Asian, Southwest Asian; Global Y-Chromosome haplogroup resolution | 163 binary SNPs<br>120 Y-SNPs | [43] |
| Precision ID Ancestry | Autosomal | African, Native American, East Asian, European, Oceanian, South Asian, Southwest Asian | 165 binary SNPs | [15,16,40,51] |
| VISAGE Basic Tool for Appearance & Ancestry# | Autosomal | African, Native American, East Asian, European, Oceanian, South Asian | 100 binary SNPs<br>15 tri-allelic SNPs | [52] |
| VISAGE Enhanced Tool for Appearance & Ancestry# | Autosomal<br>X-Chromosome<br>Y-Chromosome | African, Native American, East Asian, European, Oceanian, South Asian | 75 binary SNPs<br>26 tri-allelic SNPs<br>21 microhaplotypes<br>85 Y-SNPs<br>16 X-SNPs | [53,54] |

SNP = Single Nucleotide Polymorphism; *Only characteristics of AIMs detailed here, markers for other forensic DNA intelligence applications not shown. **Combined panel of Single Nucleotide Polymorphisms (SNPs) for identification, kinship analysis and the prediction of BGA and Externally Visible Characteristics (EVCs). ***Combined panel with Short Tandem Repeats (STRs) and SNPs for identification and SNPs for the prediction of BGA and EVCs. #Combined panel of SNPs for the prediction of BGA and EVCs.

**Table 2**
Summary of population data included in publicly accessible databases used to collate reference datasets for forensic biogeographical ancestry prediction capabilities.

| Database | Number of Individuals | Continental Populations Included | Number of Populations | Individuals per Population | Genotyping Method | Reference (s) |
|---|---|---|---|---|---|---|
| 1000 Genomes Project Phase 3 | 2054 | African, European, Central/South Asian, East Asian, Native American | 26 | 105–280 | High Coverage WGS | [57,58] |
| Estonian Biocentre Genome Diversity Panel (EGDP) | 483 | African, European, Middle Eastern, Central Asian, South Asian, East Asian, Native American, Oceanian | 148 | 1 – 17 | High Coverage WGS | [61] |
| Human Genome Diversity Project (HGDP-CEPH) | 944 | African, European, Middle Eastern, Central/South Asian, East Asian, Native American, Oceanian | 51 | 5–46 | High Coverage WGS | [59,60] |
| Simons Foundation Genome Diversity Panel (SGDP) | 260 | African, European, Central Asian, South Asian, East Asian, Native American, Oceanian | 127 | 1–15 | High Coverage WGS | [62] |

High Coverage = ≥ 30x; WGS = Whole Genome Sequencing

availability of relevant reference population data is an ongoing challenge that can cause interpretation limitations that may impact operational use. Insufficient population data for POI due to small sample size may result in inadequate representation of the genetic variation of the broader population. Consequently, this can bias BGA predictions if relied upon [64,65]. Unequal representation of relevant POI within a reference set can also contribute to bias in BGA predictions through over-estimation of admixture proportions for over-represented populations [65,66]. Operationally, this could result in incorrect conclusions being drawn and/or impact the statistical reliability of quantitative estimates accompanying a prediction. Currently, there is broad representation of major global populations across populations in geographic regions such as Sub-Saharan Africa, Europe and East Asia, and to a lesser extent, South Asia. Population data for the Middle East, North Africa, South American and Southeast Asian regions are often limited and/or less diverse [57,59]. Representation of populations across the extremely broad Oceanian region is particularly scarce and only includes populations from Papua New Guinea [59,60,62,67]. In addition, data from historically admixed populations across the Americas [68,69] are also limited across publicly available datasets [63].

At present, global population databases created specifically for forensic BGA prediction do not exist. However, the data within publicly available databases has enabled the development of forensic resources such as the Forensic MPS AIMs Panel Reference Population Sets hosted on the Snipper 4 app suite [70]. These reference datasets have been collated specifically for several of the forensic MPS AIM sets outlined in Table 1. Population samples in these data sets comprise admixed and non-admixed donors from the 1000 Genomes Project HGDP-CEPH, SGDP and EGDP data sets (Table 2). The largescale reference data sets (200k and 10k SNPs) used in the comparative study by Resutik *et al.* (2023) [71] have also been made available. Combined, these comprise data from 3957 unique individuals across the 1000 Genomes Project, HGDP-CEPH, SGDP and EDGP publicly available databases. These datasets are a valuable resource to collate population samples for AIMs not covered in the panels included in the Snipper datasets, including novel AIMs.

The release of updated high coverage WGS data for 1000 Genomes Project and HGDP-CEPH datasets has resolved prior challenges with limited coverage of some forensically relevant AIMs that resulted from a reliance on whole genome array technology for earlier releases which generated bi-allelic SNP data [63]. WGS data from the 1000 Genomes Project, HDGP-CEPH and SGDP are now curated as part of the International Genome Sample Resource (IGSR) [67] enabling reference data sets for forensic AIM panels to be reviewed and updated if required.

### 2.3. *Classification algorithms and analysis tools*

Classification algorithms, also termed classifiers, are used to generate a BGA prediction for unknown samples [5,13,72]. These classifiers rely on input of training data from known reference population

samples to define distinct groups to which unknown samples can be assigned as part of the classification process [65,73].

Broadly, classifiers used for BGA prediction can be categorised as either distance-based or model-based classifiers. Distance-based classifiers, such as multi-dimensional scaling techniques coupled with Principal Components Analysis (PCA) or Principal Co-ordinates Analysis (PCoA) to visualise results, have been commonly used in operational BGA prediction workflows, either as standalone classifiers such as in the ForenSeq Universal Analysis Software [42,74] or alongside other tools to assist interpretation [19,65]. A limitation of distance-based tools, however, is their sensitivity to sampling error (under or over-representation of a genotype). This can result in erroneous BGA predictions when the reference or training data is not suitably representative of an unknown sample [42,65]. These methods also may not provide sufficient differentiation between reference populations to enable a clear determination of BGA for unknown samples [55]. Additionally, these tools are limited in their capacity to detect and resolve admixture which leads to challenges for interpretation, particularly when relied upon as a standalone classifier [65]. Model-based classifiers, such as those using Bayesian or maximum likelihood models, offer several advantages for operational use in forensic BGA prediction methods. Bayesian algorithms have been shown to consistently outperform distance-based classifiers, providing higher accuracy when considering 'assigned' compared to 'declared' population of origin [73,75]. The accuracy remains high even with varying degrees of missing data. This is an important consideration in an operational context due to the poor quality of many forensic samples which often results in incomplete genotype data. In addition, some model-based classifiers can assist with admixture resolution. Table 3 provides a summary of the characteristics of BGA classifiers evaluated for use with forensic AIM panels.

The Snipper 4 app suite, hosted by the University of Santiago de Compostela [84], includes a Bayesian classification tool designed for forensic applications [13]. The tool provides −log likelihood values for each population group and a likelihood ratio (LR) in all BGA estimates, comparing likelihoods between the most probable and second most probable population groups [36]. Accompanying the output is a tool to enable visual assessment of the distribution of −log likelihood values between unknown and reference samples which may provide a useful approach to understand the degree of similarity (or dissimilarity) between unknown samples and reference population samples to assist interpretation [5,18]. Also included is a distance-based classification function using PCA to provide a visual display of population differentiation. It should be noted, however, that Snipper has several limitations as a standalone classifier for operational use. Currently, only binary SNP genotype data can be used, and the output provided is a categorical classification, which limits its capacity to resolve admixed BGAs [65]. LRs are assigned in all instances, including those where the true population of origin may not be represented in the reference population data. There is also limited information available to understand how the magnitude of the LR relates to the reliability or uncertainty in the result.

STRUCTURE, a Bayesian clustering tool widely used for population

**Table 3**
Classification algorithms and analysis tools evaluated for use in forensic biogeographical ancestry prediction capabilities.

| Classification tool | Classifier type(s) | Marker compatibility | Output | Suitable for resolution of admixture | Published assessment with AIM panel(s) | Original reference (s) | Independent evaluation reference(s) |
|---|---|---|---|---|---|---|---|
| ForenSeq Universal Analysis software | Principal Components Analysis (PCA) | Binary SNPs | PCA plot with centroids | No | ForenSeq SPK | | [42,74] |
| GenoGeographer | Likelihood Ratio Test | Binary SNPs | Z-scores with p-values<br>−log likelihood distribution with CIs<br>Likelihood ratios (non-excluded populations only) | Yes, first order only | PIAP<br>EUROFORGEN NAME | [46,76–78] | |
| HID Genotyper Admixture Analysis (Converge™ platform) | Maximum-Likelihood | Binary SNPs | Quantitative admixture proportions with CIs | Yes | PIAP | [79] | |
| Partial Least Squares-Discriminant Analysis (PLS-DA) | Multivariate analysis using machine learning | Binary SNPs<br>Multi-allelic SNPs<br>STRs | Categorical prediction with PLS-DA plot | No | EUROFORGEN gAIMs<br>MAPlex<br>PIAP<br>ForenSeq SPK | [55] | |
| Snipper | Naïve Bayesian and PCA | Binary SNPs | Categorical prediction with −log likelihood distribution<br>Likelihood ratio (1st/2nd population)<br>PCA plot with centroids | No | 34-plex SNPforID | [22,35,36,80] | [73] |
| STRUCTURE | Naïve Bayesian | Binary SNPs<br>Multi-allelic SNPs<br>Microhaplotypes<br>InDels<br>STRs | Quantitative admixture proportions | Yes | PIAP<br>MAPlex<br>VISAGE-BT<br>VISAGE-ET<br>EUROFORGEN gAIMs<br>EUROFORGEN NAME<br>34-plex SNPforID | [72,81] | [12,14,19,35,44–46,53,71,73,75,82] |
| Tabular Prior-data Fitted Network (TabPFN) | Tabular foundation model using neural networks | Binary SNPs<br>Multi-allelic SNPs<br>Microhaplotypes | Categorical prediction with probabilities | No | VISAGE-ET | [83] | |

SNP = Single Nucleotide Polymorphism; InDel = Insertion/Deletion polymorphism; STR = Short Tandem Repeat; PCA = Principal Components Analysis; CIs = Confidence Intervals; PIAP = Precision ID™ Ancestry Panel; MAPlex = Multiplex for the Asia-Pacific; VISAGE-BT = VISAGE Basic Tool for Ancestry & Appearance; VISAGE-ET = VISAGE Enhanced Tool for Ancestry & Appearance; NAME = North African & Middle Eastern panel; ForenSeq SPK = ForenSeq Signature Prep Kit

genetics applications has been a mainstay tool for operational BGA prediction workflows [72,82]. STRUCTURE uses a Markov Chain Monte Carlo algorithm to estimate posterior probabilities for cluster membership proportions across the 'K' defined populations in the reference dataset [65,72]. Analyses in STRUCTURE can be run with either a supervised or unsupervised approach by changing the USEPOPINFO parameter [85]. In this way, it can both be used as a classifier to assign BGA to unknown individuals based on a reference population dataset or could be used to understand population structure even when no prior information of population groups is available, or information is limited. Key advantages of STRUCTURE include the ability to utilise data from different marker types such as binary SNPs, multi-allelic SNPs and microhaplotypes in a single analysis [19,77,84]. This has become an important operational feature as forensic AIM panels continue to be expanded to include multiple marker types (Table 1). STRUCTURE, and similar clustering approaches, are also capable of resolving admixture by virtue of the quantitative output which provides the estimated cluster membership proportions corresponding to defined populations in the reference dataset [73,78,82,86]. Additionally, STRUCTURE analyses have been demonstrated to be highly robust to missing genotype data for unknown samples [73,75].

GenoGeographer is an emerging BGA prediction tool developed specifically for forensic applications by biostatisticians at the University of Copenhagen [76,77]. The tool was designed to resolve the issue of forced classification that occurs with other model-based classification tools which make an underlying assumption that the population of origin for unknown samples is included and adequately represented in the reference data [65]. GenoGeographer, in contrast, does not make this assumption, rather uses an LR test (LRT) as a measure of absolute concordance between the characteristics of the unknown sample and reference population samples. Similarly to Snipper, −log likelihoods are provided for each POI, accompanied by a confidence interval (CI). The z-scores and p-values generated from the LRT are included in the output, to indicate populations that have and have not been rejected as a population of origin at a selected confidence level. This information is used to highlight instances where the reference data does not sufficiently represent the BGA of an unknown sample being analysed and can assist operational decision-making during interpretation and reporting. An additional function includes the analysis of data with first order (family-level) admixture, where an individual's parents are assumed to have different BGAs [78]. However, the tool is not currently suitable for the analysis of more complex admixture scenarios such as admixture present in previous generations or admixture from more than two distinct populations and does not provide a quantitative output for reporting admixture proportions. The reliability of GenoGeographer for application to forensic casework has not yet been independently evaluated or compared against existing tools. Additional operational assessments of the tool would therefore be of benefit.

The HID Genotyper Admixture Analysis tool [79] embedded within the commercially available Converge™ software platform offered by Thermo Fisher Scientific (Waltham, CA) [87] replaced the HID SNP Genotyper tool previously validated and used operationally with the Precision ID Ancestry Panel (PIAP) [16,21,40]. The updated tool has been designed to assist with resolution and reporting of admixed ancestry utilising a maximum likelihood approach with bootstrapped resampling to report admixture proportions with an accompanying CI as a measure of uncertainty [79,88]. Comparable to Snipper and GenoGeographer, −log likelihoods are reported for each population group. Independent operational evaluations of the updated HID Genotyper tool have not yet been published. However, a performance assessment conducted by the software developers has been published using the default parameters and the default reference data as the training set [89] to analyse 1000 Genomes Project populations data as a test set [79]. This testing demonstrated high accuracy in BGA assignment for non-admixed African, East Asian and Northern European individuals. Variation was observed with Southern European individuals and admixed individuals. Further investigations of this classifier using modified parameters and optimised reference datasets would be valuable to determine if performance can be improved for operational use and to compare performance against established tools such as STRUCTURE and Snipper.

Recent development of novel classifiers for forensic BGA prediction capabilities have focused on machine learning (ML) algorithms that have been successfully used in other applications [90–93]. Two such ML approaches, Partial Least Squares–Discriminant Analysis (PLS-DA) [55] and Tabular Prior-data Fitted Network (TabPFN) [83] have been evaluated for use with forensic AIM panels and demonstrate similar or better prediction accuracy compared to STRUCTURE and/or Snipper in cross-validation studies. PLS-DA offers a supervised classification tool to provide a prediction based on multivariate analysis of the test and training data, providing enhancements compared to other unsupervised multidimensional scaling methods, such as PCA and PCoA [55]. TabPFN on the other hand is a pre-trained predictive model utilising transformer neural networks for the supervised analysis of tabular data to compute missing values. This approach has been applied to predicting BGA for test samples based on comparison to a tabular training set comprising AIM genotype data from reference population samples [83]. These tools are suitable for use with multiple marker types included in many of the forensic MPS AIM panels. The output from both tools is a categorical prediction of the population within the training data with the best fit to features of the unknown sample. As a result, they are not suitable for the resolution of admixture, which limits their utility as a standalone classification tool for operational use. Further evaluation of these tools with varied training and test data, including missing data and data from admixed individuals, would be beneficial to understand their utility as a tool for operational use.

Other operational tools in development include software developed by the VISAGE Consortium to support the analysis of AIM genotype data generated using the VISAGE-BT and VISAGE-ET A&A panels, signposted for publication in the near future [53]. A novel method for improved resolution of recent admixture developed by Pfaffelhuber *et al.* [94] indicated its potential suitability for operational use alongside other classifiers.

## 3. Operational considerations and challenges

The introduction of BGA prediction alongside other FDI capabilities over the past 20 years has enhanced opportunities to support investigations by expanding the value of DNA evidence. However, it also requires additional considerations for the forensic community compared to traditional DNA analysis methods using STRs [95–97]. BGA prediction capabilities have been successfully applied to casework within Australia and internationally, with operational implementation currently limited to a few jurisdictions [19–22,40]. Implementation of an operational BGA prediction capability requires an understanding of the key components of method development and optimisation, validation, the development of interpretation and reporting guidelines, as well as training programs and quality control frameworks. Fig. 1 provides an overview of the key considerations for operational implementation.

### *3.1. Defining the scope and determining critical parameters for the capability*

The first step towards operational implementation of a BGA prediction capability should be to define the scope of application and determine the critical parameters for the capability. In doing so, forensic service providers should consider how the capability will be used by LE agencies. Consultation with LE agencies at the beginning of the process is recommended to ensure the capability is designed and implemented in a way that aligns with the needs of detectives/investigators as end users. Throughout this process, determinations should be made about the POI and the level of resolution between populations that is required to provide informative results. For instance, it should be determined

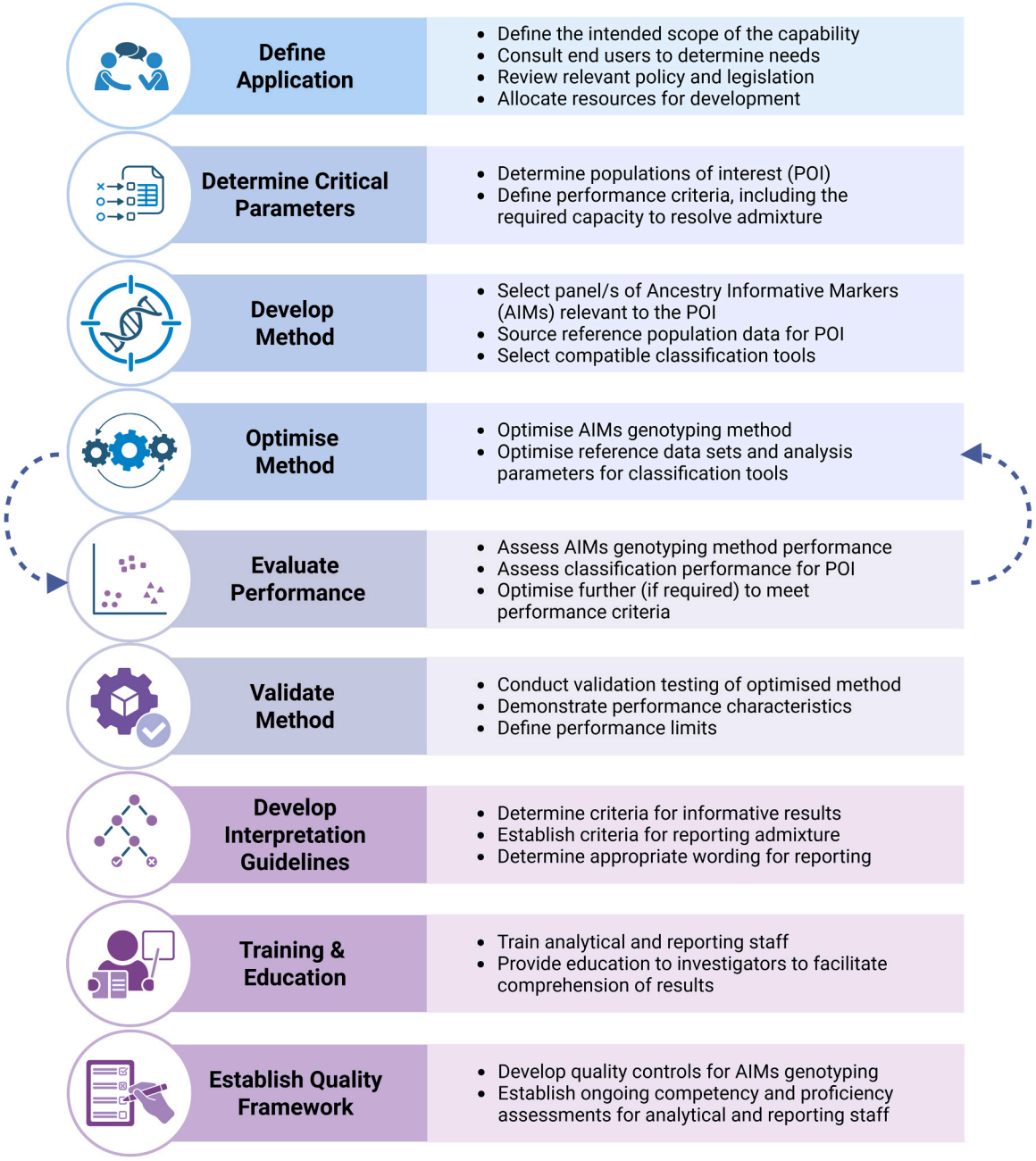



**Fig. 1.** Overview of considerations for operational implementation of a BGA prediction capability. The intended scope of the capability should be defined in consultation with law enforcement agencies, which informs key decisions about critical parameters and method development. The method requires optimisation, which may involve several rounds of evaluating performance against defined criteria and re-optimisation as necessary (represented by arrows). Validation testing is performed on the optimised method to demonstrate the performance characteristics and limits which inform the development of fit-for-purpose interpretation guidelines. The provision of training to scientific staff and education of end users should occur following method validation to allow for performance limitations of the method to be included in training/education materials. Consideration of the over-arching quality management framework is important to include measures specific to the BGA prediction capability are included to ensure the capability continues to meet operational needs. Figure created with BioRender.com.

whether differentiation of major continental groups is sufficient or if regional differentiation is also required. The required capacity to resolve admixture is an important consideration and may vary depending on the POI for the jurisdiction. Answering these questions requires an understanding of the local/regional population composition.

The implementation of new FDI capabilities requires a substantial investment of resources to develop the methodologies and expertise required to support operational use. A centre of specialisation model in which the capability is developed and maintained by a limited number of agencies may present the most efficient use of resources, with services provided to other agencies/jurisdictions by way of service level agreements. Similar approaches have been suggested for other FDI capabilities, such as FIGG [98]. Partnerships between forensic service providers and LE agencies should be considered to undertake joint funding applications to acquire resources required to develop and implement the capability.

#### 3.1.1. *Ethical, privacy and legislation considerations*

As part of defining the scope of BGA prediction capabilities, forensic service providers should review ethical and privacy policies along with legislation relevant to their jurisdiction to ensure compliance [99]. Legal and regulatory frameworks surrounding the use of DNA analysis in

investigations has continued to develop as technological capabilities have evolved [100] and the introduction of FDI capabilities has added another dimension to privacy considerations [95–97]. Undertaking a Privacy Impact Assessment (PIA) has been recommended as a means for jurisdictions adopting FDI capabilities to consider additional privacy impacts and develop appropriate policies and frameworks [101]. Guidance for conducting a PIA is provided in [102]. Privacy impacts associated with BGA prediction broadly relate to: i) data consent and privacy considerations relating to genetic data from population reference samples in databases for forensic use [103]; and ii) the manner in which BGA prediction is utilised operationally [96].

Care needs to be taken to ensure reference population data used in operational BGA prediction capabilities is limited only to publicly available data or data for which unambiguous, informed consent has been obtained [103]. Greater engagement between forensic service providers and researchers conducting large-scale population studies would also be beneficial to increase awareness of the potential forensic applications of genetic data. This could assist with ensuring transparency in the use of publicly available data for forensic applications along with appropriate informed consent. Forensic service providers should also consider legislation or policies that restrict the use of data from indigenous populations or other groups with religious or cultural sensitivities [103].

The capacity of BGA prediction to focus investigative resources has been highlighted as a benefit of the capability, however, care must be taken to ensure the privacy of individuals is not unduly impinged upon [97,104,105]. BGA is rooted in the evolutionary history of the genes associated with the AIMs being utilised and can only provide information about genetic ancestry. This is distinct from the social and cultural factors connected to ethnicity, and therefore cannot provide a prediction of race, ethnicity or cultural background [20]. However, concerns have been raised that end users could incorrectly conflate BGA with ethnicity when results are utilised in an investigation [105]. Forensic service providers and LE agencies should adopt the best practice recommendations published by the VISAGE Consortium for the socially responsible implementation of FDI capabilities [99], which outlines requirements for training and education programs for stakeholders to minimise the risks of misuse. The creation of independent oversight bodies for FDI capabilities should also be considered to ensure socially responsible use. The New Zealand Law Commission has recommended an expansion of the role of the judiciary in establishing an independent oversight committee/s to govern the use of DNA in forensic investigations [106]. Further, Scudder *et al.* [101] recommends the development of joint committees of senior staff from LE agencies and forensic service providers to review and authorise the use of predictive FDI capabilities on a case-by-case basis. The development of such oversight bodies/committees would be beneficial to minimise potential bias introduced by forensic scientists and investigators in deciding when to utilise BGA prediction, and how it is applied within an investigation to ensure the benefit to public safety is balanced with impacts to the privacy of individuals and groups in the community. These are important considerations for forensic service providers in the early stages of designing a BGA prediction capability to ensure its socially responsible use.

### 3.2. Method development

Once the scope of the capability has been defined, a method can be developed to align with the intended purpose. Key tasks during method development involve: i) the selection of a suitable AIM panel/s and an appropriate genotyping strategy, ii) the collation of representative population data and iii) the selection of appropriate classification/analysis tools.

#### 3.2.1. AIM panel selection

AIM panels should be selected to be highly informative for each of the POI. Understanding the informativeness of AIMs within available panel/s for the differentiation of the POI is therefore a fundamental task. The genotyping methodology and its compatibility with the desired AIM panel/s and other operational FDI capabilities should also be considered here. A summary of the populations that forensic MPS AIM panels are designed to differentiate and the classification tools that have been assessed with each of these panels are shown in Tables 1 and 3. Informativeness assessments should be conducted using reference population samples representative of the POI, with sample sizes sufficient to fully capture any allelic variation present [107]. Various methods are currently available for assessing AIM informativeness. The simplest metric is the absolute allele frequency differential observed between POI, referred to as delta (δ) [5,6]. Other commonly used metrics include the fixation index (Fst) [108] and Rosenberg's informativeness for assignment (*In*) [68] which incorporate genetic distance information in addition to allele frequency differences. Cheung *et al.* [27] demonstrated AIM panels selected using *In* had improved balance across populations and provided the lowest mean divergence from self-declared BGA when compared to those selected using other informativeness metrics, such as δ or Fst in isolation. Therefore, *In* is as useful metric to apply when assessing AIM informativeness. An important consideration in these assessments is the balance of informativeness between populations. Imbalanced informativeness across populations has been identified as a possible risk for bias in admixture proportions, tending to skew proportions toward populations with higher informativeness [11,18,27]. However, further research is required to establish the extent to which imbalance in AIM sets may contribute to error in BGA predictions when applied operationally. Many of the forensic MPS AIM panels (summarised in Table 1), such as the gAIMs [14], MAPlex, VISAGE-BT and VISAGE-ET, have demonstrated balanced informativeness between major continental populations. However, laboratories will need to assess informativeness for the POI relevant to their jurisdiction.

A nested approach with sequential analysis of different AIM panels or sub-sets of AIMs with different informativeness has been proposed as a strategy to maximise the resolution capacity between POI while maintaining balanced informativeness at each stage [91]. An example of a nested analysis approach using existing forensic AIM panels could involve initial analysis using a panel designed to differentiate global populations, such as the EUROFORGEN gAIMs panel. Where results of the initial analysis indicate an admixed Middle Eastern and European BGA, an AIM panel designed specifically to differentiate these populations such as EUROFORGEN NAME could be used. Therefore, it may be beneficial for laboratories to optimise and validate multiple AIM panels to allow a nested approach to differentiation of POI, if required.

#### 3.2.2. Collation of reference population data

The collation of data from representative reference population samples should occur in conjunction with the selection of AIM panel/s. Publicly available population databases described in Section 2.2 and summarised in Table 2 provide a valuable starting point for the collation of population data. Jurisdictions may seek to supplement publicly available data by sourcing data from existing published datasets through appropriate data access agreements. However, data privacy considerations associated with the use of genetic data for forensic applications remains a limiting factor [103]. In addition, forensic applications are often out of scope of the informed consent sought from participants involved with large-scale population studies which may also limit access.

Additional in-house generated population data is another option for laboratories seeking to supplement available datasets, particularly to increase sample sizes for under-represented populations, including from admixed population samples. When collecting population samples for inclusion in reference population sets, forensic practitioners must comply with best practice guidelines for conducting genomic research. Specific guidance on obtaining appropriate informed consent from donors providing samples or access to genomic data for forensic applications is available in the literature [109,110]. Frameworks for the

conduct of genomic research with indigenous and culturally sensitive populations should also be consulted by forensic practitioners planning sample collection activities [111]. Community consultation and engagement of local experts are key elements for donor recruitment activities. The needs of each community are different, therefore community partnership strategies must be culturally sensitive and tailored to each community or population. Prominent examples of frameworks designed for specific population groups include guidelines for genomic research with Australian Aboriginal and Torres Strait Islander populations [112], and Māori populations from New Zealand (Aotearoa) [112]. Practitioners seeking to engage with these populations should comply with these frameworks.

When collecting samples for in-house datasets, it is a common requirement for donors to provide BGA across four generations, including self, parents, grandparents, and great-grandparents, if known, to assist in assessing patterns of admixture before inclusion in a reference set [16]. However, the self-declared ancestry of donors may not align precisely with their genetic ancestry for several reasons: i) donors may report ancestry based on sociocultural factors such as ethnicity or religion, ii) donors may report only one of several ancestral backgrounds, which may not accurately reflect admixture, and/or iii) donors may not be aware of their genetic ancestry. Therefore, care should be taken to assess the performance characteristics of in-house collected reference population samples with the chosen AIM panel/s and classifier/s prior to inclusion in a reference set for operational use. The use of STRUCTURE with an unsupervised approach (POPFLAG=0) is recommended to verify self-declared ancestry of in-house samples and assess for unexpected admixture while also evaluating clustering performance with existing reference samples. Samples with unexpected admixture should be excluded from reference sets to minimise unnecessary complexity that can contribute to bias in predictions for model-based and machine learning classifiers [113,114]. The LRT as part of the GenoGeographer classifier may be useful as an additional tool to assess the absolute degree of similarity between in-house collected reference samples and existing reference data. The LRT in combination with the STRUCTURE output can help to inform decisions about whether to include in-house collected samples in the reference set [115].

#### 3.2.3. Selection of classification and analysis tools

Several factors must be considered when selecting appropriate classification and analysis tools. Tools should be compatible for analysis and interpretation of the selected AIM panels, considering multiple marker types if necessary and the complexity of the POI. The detection of admixture will be important for all jurisdictions, with the required capacity to resolve admixed BGAs dependent upon the intended purpose of the capability and the characteristics of the POI. The use of more than one classifier may be required to support BGA interpretation [18]. A distance-based tool may be useful in the first instance to understand how a test sample relates to reference population samples. This may indicate that admixture or BGA from an intermediate population is present which could then be further interrogated using a model-based tool that provides a quantitative assessment of admixture proportions. Further support for an interpretation could be obtained using tools that offer a categorical prediction with appropriate measures of uncertainty. GenoGeographer, which indicates whether the BGA of an unknown sample is suitably represented within the reference data, offers additional information that may assist decision making regarding reliability of a result. The use of multiple classification tools has been the approach taken by the New South Wales Police Force (Australia) for reporting FDI results, as described in [20] and [19] and has also been recommended in a recent review [18]. Consideration of local data privacy policies may also be required when selecting appropriate tools as some agencies may restrict transfer of data outside secure information technology systems used by forensic service providers and LE agencies which could limit use of tools that cannot be hosted in-house.

### 3.3. Method optimisation

The AIM genotyping method, the composition of reference population data, and the analysis parameters of classification tools are all aspects of BGA prediction methods that require optimisation to meet defined performance criteria.

#### 3.3.1. AIM genotyping optimisation

Optimisation of the genotyping method should focus on maximising genotype data from forensic samples, considering the range of sample types received in casework. Evaluations of genotyping performance have been published for the PIAP [15,16,21], MAPlex [50], NAME [44], ForenSeq SPK [49], VISAGE-BT [12,52,116–118] and VISAGE-ET [53,54] forensic MPS AIM panels using various methodologies and demonstrate successful genotyping of forensic sample types. These studies provide useful data for laboratories to benchmark against during performance assessments. Inhibitor tolerance and capacity to analyse degraded DNA are important considerations when optimising genotyping strategies. Recent research by Canele *et al.* [119] has identified successful strategies for overcoming known limitations with the inhibitor tolerance of Ion Ampliseq™ chemistry commonly used for MPS methodologies. Improved methods for human DNA enrichment have been shown to improve SNP genotyping success from severely compromised samples using hybridisation capture during MPS library preparation [120]. These strategies should be considered by forensic laboratories seeking to improve AIM genotyping success.

#### 3.3.2. Reference data set optimisation

The composition of reference data sets used to train classification tools requires careful consideration and optimisation. Under-represented populations in reference data can contribute to bias in BGA predictions, specifically in reported admixture proportions [33]. Ideally, sample sizes between POI would be balanced in a reference set, however, this is often not achieved due to limitations in available reference data [126]. The minimum sample sizes per population (or sub-population) required for effective differentiation may also depend on a range of factors, including the informativeness of the AIMs for each of the POI, performance characteristics of the classification tools in use as well as the acceptable level of confidence for predictions. In addition, the genetic diversity of POI is another factor that would impact the necessary minimum sample size. At present, there is no published methodology available to definitively determine the minimum sample size required for POI within reference data sets that considers all of these relevant factors. Assessments therefore need to be conducted by laboratories to determine whether populations within the training data can be differentiated with the available sample sizes and AIM panel/s in use. STRUCTURE or PCA can be utilised to assess population differentiation during reference set optimisation. In these assessments, sufficient population differentiation of each reference population or meta-population would be represented by a distinct cluster. Poor differentiation would require adjustments to be made to reduce the sample size of over-represented populations and/or increase under-represented populations before re-assessing differentiation.

#### 3.3.3. Classification tool optimisation

The specific parameters of the classification tool/s should also be optimised. Optimisation studies should be conducted using well-characterised reference samples across all POI to understand how different training dataset compositions and analysis parameters impact the classifier output. This approach will assist laboratories to determine whether sufficient population differentiation and prediction accuracy is achieved to meet specified performance criteria for each of the POI using the selected AIM panel/s and tools prior to progressing to validation. Optimisation studies should be conducted for each classification tool selected for operational use and include concordance between tools where multiple classifiers will be used in combination. Discrepancies

observed between different classification tools during optimisation can be valuable to indicate limitations with an AIM panel or the composition of the reference data used for one or more tools that requires further optimisation.

Area Under the Curve (AUC) metrics have been recommended to quantify performance comparisons across classifiers or different analysis parameters. AUC metrics represent the trade-off between the true positive rate and false positive rate to provide a balanced overview of performance in a single metric [121]. In a comparison between classifiers Cheung *et al.* [73,75] used a receiver operator curve (AUC-ROC) to assess overall accuracy of various classifiers, including STRUCTURE, a distance-based model, a generic Bayesian algorithm, and a multinomial logistic regression model. AUC-ROC has also been utilised by Alladio *et al.* [55] to assess STRUCTURE-like classifiers against novel machine learning approaches. Resutik *et al.* [71] used a similar approach with an area under the precision-recall (AUC-PR) curve to compare classification accuracy between different AIM sets and highlighted the benefit of AUC-PR models when dealing with imbalanced reference data sets [122]. In each of these studies, performance was assessed using a 'one vs. all others' approach. Refer to Section 2.1 for details of the AUC values achieved for the various forensic MPS AIM panels and classifiers tested in these assessments to inform benchmarks for in-house assessments. In addition to 'one vs. all others' calculations, pairwise assessments between each two-way population combination could be useful to further understand capacity to differentiate between specific POI.

Cross-validation is alternate method for assessing classifier performance which involves a subset of the reference data being assigned as 'Test/Unknown', while the remaining reference samples are assigned as the 'Training/Reference' set. It has also been widely used to assess the performance of predictive models and to conduct a controlled assessment of misclassification rates for BGA classifiers. Using a 'leave one out' approach for cross-validation has been preferred to minimise bias in the selection of the 'Test/Unknown' and 'Training/Reference' subsets as all samples in a reference set will be treated as 'Test/Unknown' in turn. More recently, a nested cross-validation approach has been proposed to improve estimates of model error [123]. A tool for cross-validation using a 'leave one out' approach implemented as part of the Snipper 4 app suite [124]. To our knowledge, a methodology for cross-validation of other BGA classifiers have not yet been developed.

A potential limitation of any classification tool is the propensity for overfitting which results in algorithms that perform well on training data but poorly on new/novel datasets. The use of separate 'training' and 'test' datasets during classifier performance assessments is therefore recommended to detect overfitting [90]. Careful selection of training data to include only high-quality data without unnecessary complexity, such as admixture, is recommended to reduce the risk of overfitting [113,114]. However, further research is needed to understand the extent to which the inclusion of admixed reference population samples contributes to overfitting with different operational BGA prediction tools.

The parameters of the STRUCTURE classification tool and their impact on results outputs have been well characterised by Porras-Hurtado *et al.* [82]. Use of the POPALPHAS parameter in STRUCTURE to calculate a separate alpha (α) value for each population has been proposed as a way to mitigate the impact of imbalances in training datasets [71]. Further investigation of the benefits of this function should be conducted as part of optimisation studies using STRUCTURE. Further studies designed to assess how different parameters affect the performance for other less-studied forensic classification tools, such as GenoGeographer, the HID Genotyper Admixture Analysis tool, as well as ML tools PLS-DA and TabPFN, would be beneficial to inform best practice.

### 3.4. Method validation

Validation of any method for forensic use is essential to demonstrate a fit-for-purpose capability and that the results can be relied upon by police and the courts [125,126]. It is also a valuable process to inform the development of suitable interpretation and reporting strategies that appropriately reflect any methodological limitations. The VISAGE Consortium has also specifically highlighted the importance of high quality validation studies as a means to ensure socially responsible use of FDI capabilities [99]. However, at present, there is an absence of best practice guidelines specific to BGA prediction to support validation efforts. Forensic DNA laboratories have instead relied on validation guidelines intended for STR analysis methods to guide BGA prediction validations, such as those published by the Scientific Working Group for DNA Analysis Methods (SWGDAM) [127]. Recommended studies on the sensitivity, robustness, reproducibility, repeatability, species specificity and performance with DNA mixtures and case-type samples are suitable when adapted to performance assessments of the chemistries and instrumentation used for genotyping AIMs. Many published validations and evaluations of forensic BGA prediction methods conducted to date have focussed on these studies [15,16,34,50,117]. However, this is only one aspect impacting the overall reliability of a BGA prediction method. Validation for operational use must demonstrate the overall performance and reliability of the method, including the selected AIMs, collated reference data sets and chosen classification tools [40]. Where multiple classifiers are being used, assessment of the performance characteristics of the different classifiers when combined should also be considered during validation to inform interpretation guidelines.

#### 3.4.1. Classification performance assessments

Rates of misclassification compared to self-declared BGA has to date been a common approach to demonstrating overall method performance during validation [11,14,33,36]. Section 3.3.3 provides details of metrics used to assess classification performance for both optimisation and validation studies. During validation it is important that classifier performance is assessed for both non-admixed and admixed BGAs and should include assessments of the repeatability and reproducibility of predictions, including of admixture proportions for tools that provide a quantitative output. The results of repeatability and reproducibility studies can help quantify noise or uncertainty in admixture proportions which is valuable to inform thresholds for reporting of non-admixed vs. admixed results.

Admixed sample data used for validation studies should include samples representative of a wide range of admixture levels in the following scenarios: family-level admixture (defined as two or more distinct ancestral lineages represented among recent ancestors) and population-level admixture (defined as apparent admixture observed for populations intermediate to populations in the reference data) [19,128,129]. One challenge in performing such assessments remains the limited availability of suitable admixed reference population samples. Simulated admixed data with known BGA proportions, as utilised by Cheung *et al.* [75] and Tvedebrink & Erikson [78], may be beneficial to supplement comparisons using self-declared data when assessing relative performance between classifiers, parameters and/or different reference data sets. Assessments of admixture resolution can compare against known admixture proportions, either based on self-declared ancestry over multiple generations or against assigned admixture proportions if using simulated data. Additionally, these assessments are valuable for the detection of apparent admixture within non-admixed POI, which could indicate that the reference data is not suitably representative or the AIM panel lacks markers informative of the POI. In these instances, laboratories should consider changes in their AIM panel selection and further optimisation of reference data sets.

A limitation of many of the available tools is the potential for classifiers to provide a prediction even where an unknown sample does not cluster well with any of the available reference populations, i.e., forced classification [65]. In an operational context this could result in BGA being assigned in instances when the population of origin of an unknown sample is not represented amongst the reference populations [76]. Given the known limitations of available reference population data, it is likely

that operational scenarios will arise in which unknown samples are not appropriately represented in reference data used for classification. These circumstances represent the potential for error using the available data and tools. Forensic laboratories validating BGA prediction capabilities should therefore assess the performance of methods using test samples that are not represented, or are under-represented, in the training data to identify when classifiers may provide unreliable or erroneous results. The use of multiple classification tools in combination may be useful to identify patterns of forced classification where discrepancies are observed in the results from different tools during validation. A recommended method for identifying the possibility of forced classification when using tools that report −log likelihoods is to plot the values for test samples against likelihoods for individuals in the reference (training) data to determine if there is sufficient overlap in the observed ranges [18]. Assessments based on this approach during validation could be useful to set minimum thresholds for −log likelihood values for each assigned BGA population. The LRT performed by the GenoGeographer tool offers another approach, clearly indicating populations can be rejected as a population of origin based on insufficient similarity [77]. Operationally, these validation assessments may be valuable to set guidelines for triaging results to identify those that are not suitable for further interpretation to avoid the reporting of erroneous results.

#### 3.4.2. Robustness

Assessing the robustness of the BGA predictions with real-world conditions is a key aspect of validation. Despite best efforts to optimise AIM genotyping methods, incomplete genotyping is likely to remain a common operational challenge due to the compromised nature of many forensic samples. Validation studies should be conducted to establish the robustness of BGA predictions with incomplete genotype data using the chosen method. This is essential to inform interpretation strategies. Studies assessing classifier performance with missing genotype data should focus on AIMs that have been shown to have poorer genotyping performance during sensitivity assessments of the AIM genotyping method. Robustness of BGA prediction methods with mixed source samples should also be considered. To date few studies have included extensive assessments of BGA prediction methods with mixed source DNA samples [16,50] with existing studies analysing a limited range of mixture ratios and contributor BGA combinations. Further evaluation and validation testing would be required to support the analysis of mixed source samples operationally. In the absence of thorough validation of the analysis of mixed source samples, laboratories should establish strategies to identify samples or results that are not suitable for further interpretation due to having a mixed DNA source.

### 3.5. Interpretation and reporting

The aim of interpretation and reporting practices are to provide investigators with a reliable result which appropriately acknowledges and explains the associated uncertainty and limitations. Currently, there are no published best practice guidelines for the interpretation and reporting of forensic BGA prediction results. However, there have been several recommendations in the literature (detailed below) which may assist laboratories in developing their own guidelines. It is important to note that laboratories will need to ensure reporting guidelines appropriately reflect the limitations of the particular methodology in use, informed by internal validation.

Key aspects of interpretation and reporting guidelines include: i) thresholds for unreliable and/or uninformative results, ii) thresholds for reporting non-admixed and admixed BGA results, which includes consideration of exclusions and measurements for associated uncertainty, and iii) report wording and structure.

#### 3.5.1. Identification of unreliable or uninformative results

Establishing clear processes for the identification of uninformative results is an important aspect of interpretation guidelines to prevent possible reporting of unreliable or erroneous results. A decision-tree model has been developed to assist in the reporting of BGA prediction results as part of the operational BGA capability developed for the Australian Defence Force's Unrecovered War Casualties – Army (UWC-A) to support investigations conducted to identify historical military remains [129]. A similar decision-tree approach could be implemented by any jurisdiction implementing a BGA prediction capability to support operational decision making. In addition, the input parameters for the tree could be tailored to suit the investigative question being considered.

#### 3.5.2. Reporting non-admixed and admixed results

The capacity to interpret and reliably report admixture requires careful consideration. Thresholds to determine non-admixed compared to admixed results will be method dependent and should be informed by validation data. In some instances, admixed ancestry results may be deemed uninformative if the admixture cannot be reliably resolved. For example, it is currently not possible to differentiate family-level admixture from population-level admixture with available tools. McNevin [19] advocates for all possibilities to be reported where BGA results can be explained by different genetic mechanisms. However, these descriptions may not be operationally informative particularly if being provided to non-scientific audiences such as detectives or investigators. Further discussion with LE agencies as end users would be beneficial to determine appropriate guidelines for uninformative results when considering admixture. Refer to Section 3.6 for training and education considerations.

#### 3.5.3. Exclusions

One aspect not explicitly covered in the literature is the reporting of exclusions. Providing investigators with the likely BGA of an unknown donor is a primary goal, however, the ability to exclude other possible populations as contributing to BGA also provides valuable intelligence to guide an investigation. Operational validation studies and reporting guidelines should consider how exclusions may be incorporated. GenoGeographer, in its capacity to assess concordance between unknown samples and the available reference populations, may be utilised to aid in determining which populations can be reliably excluded [76,77].

#### 3.5.4. Measures of uncertainty

Uncertainty is expected in all BGA prediction results and the inclusion of measures of uncertainty are useful to support decision-making when reporting and utilising BGA prediction results in an operational context. LRs have been used to provide a measure of the weight of evidence in forensic DNA applications such as STR typing [130,131], and have been recommended for BGA prediction [13,36]. LRs relate only to a binary scenario which requires careful application when attempting to answer the question of 'what is an unknown individual's BGA?', as this is not a binary question. However, with a sequential approach and tools to reliably exclude populations, a binary question may be derived. The Snipper and GenoGeographer classification tools both include functions for the calculation and reporting of LRs, however, the approach to reporting LRs varies between these tools. Snipper provides an LR comparison of the two populations with highest likelihoods, regardless of the values. In contrast GenoGeographer, is more discriminatory and will only provide LRs for populations which have not been rejected following the LRT for absolute concordance.

Other measures of uncertainty such as CIs have also been recommended to support results interpretation in operational contexts [129] and have been incorporated into the output of some BGA classifiers. The HID Genotyper Admixture Analysis function incorporates CIs based on bootstrap resampling variation [79] and GenoGeographer includes CIs for −log likelihood estimates. Further research to assess the reliability of LRs, CIs and/or other measures of uncertainty, along with their potential benefit in assisting BGA interpretation is needed to develop guidance for best practice. Consideration should also be given to how measures of uncertainty could be incorporated for reporting purposes. Caution is

required to ensure that measures are appropriately evaluated by investigators, particularly in the case of LRs where investigators may be familiar with the magnitude of LR values provided for sub-source reporting of STR results.

#### 3.5.5. Results communication

The language or wording used to describe, or communicate, the outcomes of BGA analysis is a key consideration. As demonstrated in the outcomes of a provider trial published by Atwood *et al.* [20] there can be substantial variation in the wording and information provided alongside BGA predictions in reports provided to LE agencies. The authors highlight that this variability in nomenclature could contribute to misunderstanding and potential erroneous application of results in investigations. Efforts to standardise nomenclature are required to support results comprehension. "Majority ancestral contribution from [population]" has been proposed as wording for reporting non-admixed results to allow for uncertainty in the prediction, an improvement compared to other approaches listing only the population deemed most likely [20]. BGA prediction outputs are directly related to the reference population samples utilised in the classification process in conjunction with the informativeness of the selected AIM panel. It has therefore been recommended that a description of reference population samples be included in reports alongside predictions [18]. Such information could provide valuable supporting information for predictions and exclusions while also acknowledging any deficiencies in representation.

In addition, consideration is required to determine the most appropriate wording to report admixed results which can represent the broad range of admixture scenarios likely to be encountered in casework. Jin *et al.* [21] outlined reporting guidelines based on an operational validation of the PIAP AIM panel and HID SNP Genotyper classifier (now superseded by HID Genotyper Admixture Analysis Tool). In these guidelines, predictions are reported as one of four categories based on observed admixture proportions: *single inclusion*, *single mixed inclusion*, *double inclusion* and *uninformative result*. While the exact admixture proportions and wording chosen for the categories detailed by the authors are specific to their methodology and validation, the clear definitions incorporating quantitative information for non-admixed and reportable admixed results, as well as uninformative results provides a framework for others in developing operational guidelines.

### 3.6. Training and education

Training and education of interpretation and reporting scientists as well as detectives and investigators is imperative to ensure accuracy in the delivery and application of the BGA results. Development of relevant training and education programs are required for all forensic practitioners involved in generating, interpreting and reporting BGA predictions, as well as for police and investigators as those who receive the investigative lead to ensure results are understood and applied appropriately. Implementation of BGA prediction capabilities within Australian jurisdictions to date has involved a considered approach to the delivery of BGA prediction results to investigators which has focused on the provision of education programs to assist comprehension of BGA. Result provision in this context has been accompanied by a training session with investigators to clearly define what the BGA prediction capability is, and is not, providing [20,99]. In addition, the New South Wales Police Force approach to BGA training and education involved engaging investigators with BGA output from known donors, providing education relating to non-admixed and admixed individuals in the context of genetic ancestry (as opposed to ethnicity) and communicating the limitations of BGA analysis. In addition, case reviews were conducted with investigators who had received prior BGA training. This aligns with recommendations from the VISAGE Consortium for the socially responsible use of BGA prediction [99]. Further collaboration between forensic laboratories would be beneficial to develop accepted guidelines for training programs for scientists and investigators.

### 3.7. Quality management

Quality management is critical in the implementation of a new capability. Method validation, training and education programs as discussed above, are important aspects of quality assurance practices [132]. However, consideration must also be given to the development of appropriate quality control measures specific to BGA prediction methods for ongoing quality management. The use of appropriate quality control materials is required for monitoring the quality of AIM genotyping processes. This includes the use of a reagent blanks or other no template controls to monitor for contamination throughout analytical processes. Inclusion of a well-characterised known reference sample alongside test samples is also recommended for forensic genetic testing purposes to ensure analytical methods are performing as intended [133]. Commercial genetic reference materials developed for whole genome sequencing applications, which have verified BGA, are available for purchase from NIST (National Institute of Standards & Technology, United States) [134]. These provide a valuable resource for laboratories monitoring AIM genotyping performance. Consistent with requirements for routine forensic analyses, technical review processes for the analysis, interpretation and reporting of BGA prediction results should be implemented [132], and where possible these reviews should be conducted blind, without awareness of the findings of other analysts or reporters [135]. At present, commercial proficiency testing programs are not available for BGA prediction capabilities. In the absence of formal proficiency testing programs, laboratories should seek to engage in inter-laboratory comparisons as a quality control measure. Inter-laboratory evaluations published by the EUROFORGEN and EDNAP groups may provide guidance to other jurisdictions seeking to create and engage in similar inter-laboratory comparisons [34,45].

## 4. Conclusion

BGA prediction will remain a valuable FDI tool to assist investigations through the provision of additional information about the donor of biological evidence. Further research to improve and standardise processes for method optimisation, validation, interpretation and reporting of results as well as training and education programs would be highly beneficial to enhance consistency in validation and application of BGA prediction capabilities. Continued communication between forensic service providers and LE agencies during this process is encouraged to ensure alignment between the intended use and the performance characteristics of the capability, essential to the socially responsible implementation of BGA prediction.

## CRediT authorship contribution statement

**Felicity Poulsen:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Conceptualization. **Lyn Griffiths:** Writing – review & editing, Supervision. **Larisa Haupt:** Writing – review & editing, Supervision. **Catherine Hitchcock:** Writing – review & editing, Supervision. **Kirsty Wright:** Writing – review & editing, Supervision, Conceptualization. **Runa Daniel:** Writing – review & editing, Supervision, Conceptualization.

## Declaration of Competing Interest

The authors declare no conflicts of interest.

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